1. Understand and utilise constructs based on definitions, rules and formal systems including employing familiar algorithms (Point 5 in the list)

This is the core ability required to solve this problem. Students need to follow basic geometric rules and the definition of a scale to calculate length.

- **Specific Analysis:** Students must understand and apply the rule of the scale: one unit length on the grid represents 0.5 meters in the real world. They need to identify that the counter edge consists of horizontal, vertical, and diagonal line segments, calculate or measure the length of these segments on the grid, and then apply the scale for conversion. This is a direct application of established mathematical rules and algorithms (length calculation).

2. Use a multi-step procedure leading to a mathematical solution, conclusion or generalisation (Point 12 in the list)

Calculating the total length requires multiple steps.

- **Specific Analysis:**
  1. Determine the grid length of each line segment.
  2. For diagonal segments, students might need to apply the Pythagorean theorem (although recognizing the diagonal of a square can simplify it).
  3. Add all the grid lengths of the segments to get the total grid length.
  4. Multiply the total grid length by 0.5 meters (scale conversion).
     This is a clear multi-step procedure.

3. Manipulate numbers, graphical and statistical data and information, algebraic expressions and equations, and geometric representations (Point 7 in the list)

This task requires students to manipulate multiple types of information.

- **Specific Analysis:** Students need to extract information from a graphical representation (floor plan), manipulate numbers (calculate lengths and apply the scale), and represent the result as a final length value. This involves switching between different representations and manipulating data.

4. Articulate a solution, showing and/or summarising and presenting intermediate mathematical results (Point 8 in the list)

The question explicitly asks to "Show your work".

- **Specific Analysis:** Students must provide not just the final answer, but also clearly display how they arrived at the length of each segment and the calculation process for the total length. This emphasizes the ability to communicate and document mathematical reasoning.

1. 利用基于定义、规则和形式系统的构建，包括运用熟悉的算法 (列表中的第 5 点)

这是解决此问题的核心能力。学生需要遵循基本的几何规则和比例尺定义来计算长度。

- **具体分析：** 学生必须理解并应用比例尺的规则：网格上的一个单位长度代表现实世界中的 0.5 米。他们需要识别出柜台边缘由水平线、垂直线和对角线段组成，并通过计算或测量这些线段在网格上的长度，然后应用比例尺进行转换。这是对既定数学规则和算法（长度计算）的直接应用。

2. 利用多步骤程序得出数学解、结论或概括 (列表中的第 12 点)

计算总长度需要多个步骤。

- **具体分析：**
  1. 确定每个线段的网格长度。
  2. 对于对角线段，学生可能需要应用勾股定理（尽管通过识别正方形对角线可以简化）。
  3. 将所有线段的网格长度相加得到总网格长度。
  4. 将总网格长度乘以 0.5 米（比例尺转换）。
     这是一个清晰的多步骤过程。

3. 操作数字、图形和统计数据及信息、代数表达式和方程式以及几何表示 (列表中的第 7 点)

该任务要求学生操作多种类型的信息。

- **具体分析：** 学生需要从图形表示（平面图）中提取信息，操作数字（计算长度和应用比例尺），并将结果表示为最终的长度数值。这涉及在不同表示形式之间切换和操作数据。

4. 阐明解决方案，展示和/或总结并呈现中间数学结果 (列表中的第 8 点)

题目明确要求“Show your work”（展示你的工作）。

- **具体分析：** 学生不仅要提供最终答案，还需要清晰地展示他们是如何得出每个线段的长度以及总长度的计算过程。这强调了沟通和记录数学推理的能力。